TEERATHAM (TJ) VITCHUTRIPOP

(804)-928-4949 | tj.vitchutripop@yale.edu | tjvitchutripop.github.io

51 Prospect St, New Haven, CT 06511

RESEARCH MISSION

My research focuses on how temporally-grounded experiential and interactive learning can be realized in embodied computational systems. I am interested in designing machine learning agents that are, not only adept at processing the static content of experience, but also how it unravels across a continuous unstructured temporal axis. This effort involves representing moments and intervals of time and forging effective compositional, hierarchical abstractions over time. Thus, my work is closely aligned to the domains of continual learning and reinforcement learning with concrete applications to robotics. My research is motivated by the end-goal of developing robotic systems that can meaningfully learn and adapt over longitudinal deployment in human-centric environments (e.g., home and assistive robotics) and operate efficiently in real-time critical scenarios (e.g., surgery, disaster response).

EDUCATION

Yale University — New Haven, CT
Ph.D. in Computer Science — 2024 – Present
Advisor: Daniel Rakita

University of Virginia — Charlottesville, VA
B.S. Computer Science and B.A. Philosophy — 2020 – 2024
GPA: 3.885 | *Honors*: Highest Distinction, Raven Society

RESEARCH EXPERIENCE

Yale University, Department of Computer Science — New Haven, CT
Graduate Research Assistant – Applied Planning, Learning, and Optimization (APOLLO) Lab — Aug. 2024 – Present
Advised by **Daniel Raktia**

- Designing benchmarks and evaluation methods for continual lifelong learning in dynamic environments that require in-situ learning and adaptation at multiple timescales.
- Proposing and developing learning algorithms for effective representations and abstractions in continuous unstructured time through a single egocentric stream of experience.
- Analyzing optimizers and techniques for combatting plasticity loss in neural networks as a pathway for robotic adaptation via continual reinforcement learning.
- Brainstorming computer vision approaches for 3D view reconstruction to assist surgeons in robotic surgery.

Carnegie Mellon University, Robotics Institute — Pittsburgh, PA
RI Summer Scholar (RISS) – Robots Perceiving and Doing (R-PAD) Lab — Jun. 2023 – June 2024
Advised by **David Held**

- Proposed and developed novel unsupervised architecture, *TaskSeg*, for segmenting task-relevant objects in robot manipulation tasks through video demonstrations.
- Applied optical flow on video demonstration frames to generate pseudo-label masks used to train a segmentation model for a downstream robot manipulation policy.
- Performed comparative experiments with a model trained on ground truth data, showing comparable results (~5% mIoU difference on most tasks), and ablation studies on different flow aggregation methods.

University of Virginia, Link Lab — Charlottesville, VA
Undergraduate Research Assistant – Collaborative Robotics Lab — Aug. 2021 – May 2024
Advised by **Tariq Iqbal**

- Proposed and developed novel deep reinforcement learning algorithm, *LASSO*, to tackle dynamic goal manipulation tasks using an autoencoder and contrastive learning-based architecture, addressing the representation learning bottleneck of RL algorithms and improving upon state-of-the-art performance.
- Conducted experiments in custom OpenAI Gym MuJoCo environments to benchmark task performance.
- Developed behavior trees in ROS using PyTrees for robotic control in human-robot demonstrations.

PUBLICATIONS

2026

T. Vitchutripop, A. Quarles, W. Zhang, D. Rakita, *Stop Spatializing Time: Machine Learning Agents Should Learn Through Time, Not About Time* (Preprint — To be submitted to ICML 2027) [paper]

T. Vitchutripop, A. Quarles, W. Zhang, R. Xue, D. Rakita, *An Analysis of Streaming Deep Reinforcement Learning for Adaptive Continual Learning in Robotics* (In Submission)

H. You, Y. Liu, D. Zong, Q. Wang, **T. Vitchutripop**, Q. Wang, D. Rakita, I. Abraham, *Efficient On-policy Visual-RL via Stochastic Decoupled Policy Gradient.* (In Submission) [paper]

J. Bader, X. Sun, T. Rosenfelt, A. Ramirez-Hardy, **T. Vitchutripop**, H. Pantel, A. Khanna, and D. Rakita, *Deep learning approach for critical exposure during division of the inferior mesenteric artery in colorectal surgery*, Journal of Robotic Surgery [paper]

J. Bader, X. Sun, T. Rosenfelt, A. Ramirez-Hardy, N. Sapir, R. Scheub, **T. Vitchutripop**, A. Khanna, H. Pantel, and D. Rakita, *AI-Powered Semantic Segmentation Model for Enhanced Ureteral Mapping and Real-Time Instrument Feedback in Robotic Colorectal Surgery*, Journal of Robotic Surgery

2024

B. Eisner, E. Cai, O. Donca, **T. Vitchutripop**, and D. Held, *Sequential Object-Centric Relative Placement Prediction for Long-Horizon Imitation Learning*, Learning Effective Abstractions for Planning (LEAP) Workshop @ Conference on Robot Learning 2024 [paper]

J. Brown*, **T. Vitchutripop***, E. Cai, J. Wang, and D. Held, *Unsupervised Deep Instruction Tuning for Few Shot Object Segmentation* (Submitted) [website]

M. S. Yasar, **T. Vitchutripop**, and T. Iqbal, *LASSO: Learning Latent Policies via State Space Modeling* (Submitted)

2023

T. Vitchutripop, J. Wang, and D. Held, *TaskSeg: Task-Specific Object Segmentation Through Demonstration*, RISS Working Papers Journal 2023 [paper] [video] [poster]

PRESENTATIONS

TaskSeg: Task-Specific Object Segmentation Through Demonstration [poster] — August 2023
Poster Presentation, Robotics Institute Summer Scholar Showcase, Carnegie Mellon University

LASSO: Learning Latent Policies via State Space Modeling — April 2023
Oral Presentation, Undergraduate Engineering Research and Design Symposium, University of Virginia (Awarded Best Oral Presentation)

LASSO: Learning Latent Policies via State Space Modeling — March 2023
ACC Meeting of the Minds Conference, Virginia Tech (1 of only 5 selected to represent UVA)

HONORS & GRANTS

2025 Design and Technology Fellow — 2025
Fellowships at Auschwitz for the Study of Professional Ethics (FASPE)

Louis T. Rader Outstanding Undergraduate Research Award — 2024, 2023
Department of Computer Science, University of Virginia

CRA 2024 Outstanding Undergraduate Researcher Award (*Honorable Mention*) — 2024
Computing Research Association

Robotics Institute Summer Scholar (RISS) [NSF REU Program] *(7.8% acceptance rate)* — 2023
Robotics Institute, Carnegie Mellon University

Best Oral Presentation *(1st place)* — 2023
2023 Undergraduate Engineering Research and Design Symposium, University of Virginia

Double Hoo Research Grant Award 2022
Office of Citizen Scholar Development, University of Virginia

SKILLS & LANGUAGES

Programming Languages: Python, Java, C++, C, JavaScript, TypeScript, Assembly
Machine Learning and Robotics Frameworks: PyTorch, TensorFlow & Keras, Gymnasium, MuJoCo, ManiSkill3, ROS, ROS 2, OpenCV, Scikit-Learn, NumPy, Pandas, PyTrees, PyTorch Lightning
Other Tools and Frameworks: GitHub, Bitbucket, Docker, Weights and Biases, Singularity, Slurm, Visual Studio Code, JupyterLab, Linux, React, Node.js, Airtable, Excel, MATLAB, Autodesk Fusion 360

PROFESSIONAL EXPERIENCE

National Science Foundation Alexandria, VA
Policy and Data Science Intern – UVA-MIT Policy Internship Program June 2022 – Feb 2023

- Contributed towards efforts to publish and open-source innovation and entrepreneurship application data for the NSF Engines program, developing data cleaning pipelines, data visualization prototypes, and a public-facing database for collaboration in Airtable used by 5000+ users and featured in multiple publications (e.g., Forbes, Heartland Forward, SSTI).
- Leveraged state-of-the-art large language models and natural language processing techniques to extract entities from records and reports, unveiling companies/startups spun off from NSF-funded research.

Interop.io (*formerly Cosaic*) Charlottesville, VA
Software Engineering Intern June 2021 – Aug. 2021

- Designed headless UI unit tests for React components (increasing coverage from 0% to 50%) and end-to-end regression tests for 2 different parts of the product.
- Refactored existing legacy React components, converting them to TypeScript for build-time type safety and importing them into Storybook to support modular testing.

TEACHING EXPERIENCE

CPSC 1700 AI for Future Presidents, Yale University New Haven, CT
Graduate Teaching Fellow Jan. 2026 – May 2026

- Organized weekly discussion sections on the technical foundations and sociotechnical impact of modern AI to a non-technical introductory-level audience of 15+ students.
- Provided detailed feedback on student papers and graded assignments.

CS 2120 Discrete Mathematics and Theory 1, University of Virginia Charlottesville, VA
Teaching Assistant Feb. 2021 – May 2024

- Planned and co-lectured classes on quantifier logic and entailment to 100+ students.
- Guide and support students on course content during in-class activities, office hours, and after lectures.
- Strategize with professors and other teaching assistants about optimal ways to deliver class content.

STS 3020 Science and Technology Policy for Interns, University of Virginia Charlottesville, VA
Teaching Assistant Aug. 2022 – May 2023

- Supported instructor in program recruitment and course design & operations.
- Coordinated and moderated alumni guest speaker panels.
- Developed and maintained UVA Policy Internship Program website.

LEADERSHIP & SERVICE

Yale Pathways to Science, Yale University New Haven, CT
Volunteer Workshop Lead July 2025, 2026

- Led multiple 1-hour workshops on robotics and artificial intelligence to 40+ local high school students.
- Designed lecture material, interactive activities, and robotic demonstrations for workshop.

Girls' Science Investigation, Yale University New Haven, CT
Volunteer Event Facilitator Feb. 2025
- Assisted in setting up event and helping local middle school students complete hands-on science activities.

HooHacks, University of Virginia Charlottesville, VA
Marketing Committee Co-Chair Sept. 2020 – May 2024
- Lead committee members and collaborate with HooHacks executive board on planning marketing campaign and strategy for HooHacks, UVA's premier student-run hackathon with 1000+ participants.
- Established stronger relationships with organizations for underrepresented groups in STEM and minority serving institutions to make events more inclusive.

Charlottesville Debate League (CDL), University of Virginia Charlottesville, VA
Teacher (2020-2023) | Head Teacher (2022) Sept. 2020 – May 2023
- Mentored 30+ middle school students on extemporaneous speaking and public forum debate.
- Discuss with teachers on best ways to implement curriculum and maintain high student engagement.
- Analyze effective teaching strategies with other CDL teachers at 10+ schools.